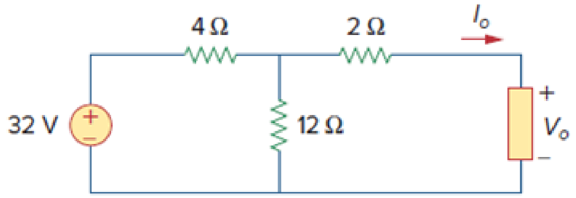
Chapter 4, Problem 65P

Problem

For the circuit shown in Fig. 4.131, determine the relationship between *Vo* and *Io*.

Figure. 4.131:



Step-by-step solution

Step 1 of 3

At the terminals of the unknown resistance, we replace the circuit by its

Thevenin's equivalent,

$$R_{Th} = 2\ \Omega + \left(4\ \Omega \| \Omega 12\right)$$

$$R_{Th} = 2 + \frac{4 \times 12}{4+12}$$

$$R_{Th} = 2\ +\ 3$$

$$R_{Th} = 5\Omega$$

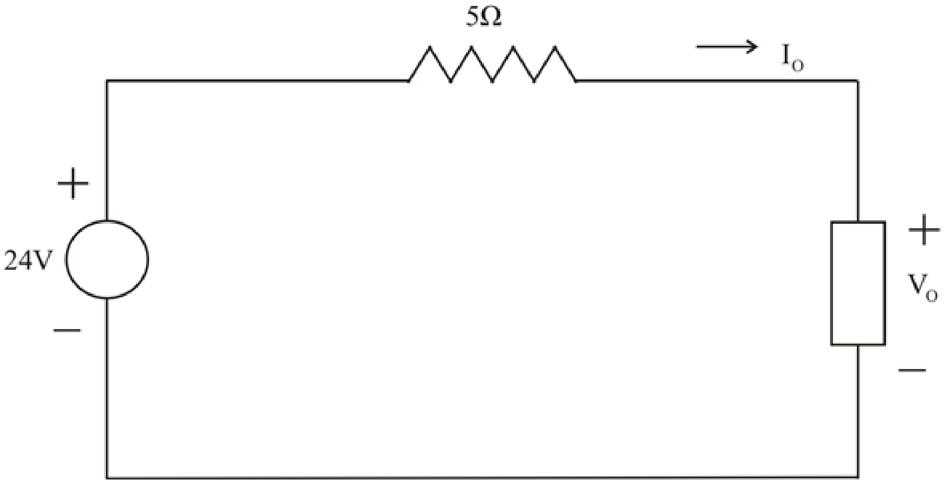
Therefore;

$$V_{Th} = \left(\frac{12}{12+4}\right) \times 32$$

$$\boxed{V_{Th} = 24\,\text{V}}$$

Thus, the circuit can be replaced by that shown below.

Step 2 of 3



Step 3 of 3

Applying KVL to the loop

$$-24 + 5I_0 + V_0 = 0$$

$$V_0 = 24 - 5I_0$$